Construction Chemicals

SUBMITTAL DOCUMENT

O Poxyguard

Potable Water Tank
Epoxy Lining System



Potable Water
Tank lining

Solvent-Free
Body & top coats

3 Layers
Primer – body – top

Holiday-Free
Tested lining

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

Construction Chemicals

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METHOD STATEMENT

O Poxyguard — Potable Water Tank Epoxy Lining System

Project Name		**Document No.**	OMX-SMS-02
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
System	O Poxyguard	**Status**	For Approval

1. Purpose

This method statement defines the sequence, resources, inspection and test plan and safety measures for the installation of the O Poxyguard system on site. It describes how each layer of the system is prepared, mixed, applied, inspected and protected so that the completed system achieves the performance stated in the O MIX Technical Data Sheets of its components.

2. Scope of Work

O Poxyguard is a solvent-free epoxy lining system for potable water tanks and reservoirs. A surface-tolerant epoxy primer is followed by body and top coats of O-Guard EP, a polyamine epoxy suitable for contact with drinking water and food.

- Substrate inspection, testing and acceptance.
- Surface preparation and repair of defects.
- Stage 1: Primer — O-Prime ST.
- Stage 2: Body coat — O-Guard EP.
- Stage 3: Top coat — O-Guard EP.
- Curing, protection, final testing and handover.

2.1 System build-up

Stage	Product	Function	Consumption
1	O-Prime ST	Primer	0.18 – 0.20 kg/m²
2	O-Guard EP	Body coat	0.25 – 0.30 kg/m²
3	O-Guard EP	Top coat	0.25 – 0.30 kg/m²

2.2 Typical areas of use

- Potable water tanks and reservoirs (concrete or steel).
- Water towers.
- Water treatment plants.
- Industrial water tanks.

3. Definitions

Term	Meaning
System	All layers of the build-up applied in the stated sequence.
Hold point	Stage at which work stops until QA/QC (and the consultant, where required) approve.
Recoat window	Minimum and maximum time between two layers for proper inter-layer bonding.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Mock-up / trial area	Area applied before the main works to approve materials, method and appearance.
Pot life	Time a mixed multi-component material remains workable.
Holiday	Pinhole or discontinuity in a lining, detected with a holiday detector.
WFT / DFT	Wet film thickness during application / dry film thickness after cure.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Daily supervision; confirms substrate readiness, climate, mixing, consumption and recoat windows; keeps daily records.
QA/QC Inspector	Carries out tests at each hold point, records results and signs the inspection checklist.
HSE Officer	Toolbox talks, PPE, ventilation, safe access and storage; stops unsafe work.
Applicator Foreman	Leads the application team and follows this method statement; reports deviations.
O MIX Technical Representative	Technical support, site visits and advice on product use when requested.

5. Materials

Product	Description	Packaging
O-Prime ST	Surface-Tolerant Epoxy Primer for Steel, Light Metals & Concrete	20 kg sets (Part A 16 kg + Part B 4 kg) and 5 kg sets (Part A 4 kg + Part B 1 kg) in metal pails.
O-Guard EP	Solvent-Free Polyamine Epoxy Lining for Potable Water Tanks	5 kg sets: Part A 4.35 kg in pail + Part B 0.65 kg in jerrycan.

5.1 Key product data

Product	Mixing ratio	Pot life / working time	Recoat / drying
O-Prime ST	4 : 1 by weight	2 hours	As per O-Prime ST recoat window
O-Guard EP	200 : 30 by weight — pre-weighed sets, mix as recommended	40 min	24 hours (max. 48 hours)

6. Tools & Equipment

- Diamond grinder and/or shot blaster with dust extraction; industrial vacuum
- Slow-speed drill mixer (300 – 500 rpm) with spiral paddle
- Rollers, brushes, squeegees, notched trowels and steel trowels as required by each stage
- Scales and measuring containers for part mixing (where permitted)
- Thermo-hygrometer, surface thermometer and dew-point calculator
- Moisture meter and plastic sheets for moisture testing
- Masking tape, protective sheeting, barriers and warning signs
- Lighting and extension cables with residual current protection
- Wet-film comb gauge
- Pull-off adhesion tester
- Spiked roller and spiked shoes
- Holiday detector (wet sponge or high-voltage spark tester)
- Ultrasonic / magnetic DFT gauge
- Airless spray pump where specified
- Forced ventilation and gas detector for confined spaces

6.1 Personal protective equipment

- Safety shoes, helmet and high-visibility vest
- Chemical-resistant gloves (nitrile) and safety goggles
- Coveralls with long sleeves
- Dust masks (P2) during grinding and powder mixing
- Respirators with organic vapour cartridges for solvent-containing products and confined spaces
- Knee pads for floor work

7. Storage & Handling

- Store all products in original, unopened packaging in a dry, shaded and ventilated store at the temperatures stated in each TDS.
- Keep Part A, Part B (and Part C) of each kit together; do not mix components from different batches.
- Protect liquids from frost and direct sun, and powders from moisture.
- Store solvent-containing products away from heat, sparks and ignition sources.
- Use materials on a first-in, first-out basis and record batch numbers for each area.
- Condition materials to +15 °C to +25 °C for 24 hours before use in hot or cold weather.

8. Pre-Start Requirements

- Approved material submittal, this method statement and the inspection and test plan.
- Approved drawings showing areas, joints, upstands, drains and details.
- Work area handed over clean, free of other trades, with safe access, lighting and power.

- Mock-up / trial area executed and approved for system, colour and finish.
- Toolbox talk held with the team on this method statement and HSE measures.
- Climate monitoring equipment calibrated and available.

9. Substrate Inspection & Acceptance (Hold Point)

Check	Acceptance criteria	Method
Concrete age	≥ 28 days (cement-based substrates)	Site records
Compressive strength	≥ 25 N/mm²	Rebound hammer (ASTM C805) or cores
Surface tensile (pull-off) strength	≥ 1.5 N/mm²	Pull-off test ASTM D7234
Moisture content	< 4 % (no moisture under plastic sheet after 16 h)	Moisture meter; plastic sheet test ASTM D4263
Steel surface cleanliness	Sa 2½ (St 2 minimum where surface-tolerant primer is accepted)	ISO 8501-1
Dust on steel	Rating ≤ 2	ISO 8502-3 tape test
Climate	Substrate ≥ 3 °C above dew point; RH ≤ 85 %	ISO 8502-4 / thermo-hygrometer

10. Surface Preparation

1. Remove oil and grease by solvent cleaning (SSPC-SP 1).
2. Abrasive blast-clean steel to Sa 2½ (or power-tool clean to St 2 where accepted for the surface-tolerant primer).
3. Remove dust; check dust rating and soluble salts where required.
4. Concrete: shot blast or diamond grind to remove laitance and open the pores.
5. Mask and protect adjacent surfaces, drains and fixtures.
6. Record substrate test results and obtain QA/QC approval before priming (hold point).

11. Application Sequence

Apply the system stage by stage in the order below. Each stage must be inspected and approved before the next stage starts, and the next stage must be applied within the recoat window stated.

11.1 Stage 1 — Primer: O-Prime ST

Consumption: 0.18 – 0.20 kg/m². Time before next stage: As per O-Prime ST recoat window.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Stir Part A thoroughly to disperse any settled pigment.
3. Add the complete Part B to Part A (4 : 1 by weight).
4. Mix 2 – 3 minutes with a slow-speed mixer until uniform in colour.
5. Add epoxy thinner only where needed for spray application, within the recommended limit.
6. Apply a stripe coat by brush on welds, edges, bolts and corners.

7. Apply the full coat by brush, roller or airless spray at 0.18 – 0.20 kg/m² (100 µm DFT) or to the DFT specified for the system.
8. Measure wet film thickness continuously and adjust application to reach the specified DFT.
9. Apply additional coats where the specified total DFT exceeds the single-coat thickness.

11.2 Stage 2 — Body coat: O-Guard EP

Consumption: 0.25 – 0.30 kg/m². Time before next stage: Min. 24 h, max. 48 h.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Add Part B to Part A (200 : 30 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. Apply the body coat by roller or airless spray at 0.25 – 0.30 kg/m².
4. Apply the top coat after 24 hours and within 48 hours.
5. Check wet film thickness during application.

11.3 Stage 3 — Top coat: O-Guard EP

Consumption: 0.25 – 0.30 kg/m². Time before next stage: Full cure 7 days; wash before filling.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Add Part B to Part A (200 : 30 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. Apply the body coat by roller or airless spray at 0.25 – 0.30 kg/m².
4. Apply the top coat after 24 hours and within 48 hours.
5. Check wet film thickness during application.

12. Hold Points & Recoat Windows

Stage	Before starting	Waiting time before next stage	Approved by
1. O-Prime ST	Substrate accepted (Section 9)	As per O-Prime ST recoat window	QA/QC
2. O-Guard EP	Previous stage inspected and approved	Min. 24 h, max. 48 h	QA/QC
3. O-Guard EP	Previous stage inspected and approved	Full cure 7 days; wash before filling	QA/QC

13. Curing, Protection & Return to Service

- Protect each layer from rain, dew, dust, water and traffic until it has cured.
- Barricade the area and display warning signs during application and curing.
- Keep ventilation running in enclosed areas until full cure.
- Do not fill or immerse the lining before full cure (7 days at 25 °C).
- Before commissioning, wash the tank with clean water and disinfect it in accordance with the local authority requirements (e.g. AWWA C652), then flush before filling.

14. Quality Control & Inspection Test Plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point (ISO 8502-4)	Start of shift and every 4 h	Daily climate log
Substrate	Strength, pull-off (ASTM D7234), moisture (ASTM D4263)	Each area (hold point)	Inspection request
Mixing	Complete kits, ratio, mixing time	Each mix	Daily record
Film thickness	WFT comb gauge (ASTM D4414) during application	Each coat	Daily record
DFT	Steel: SSPC-PA 2; concrete: ultrasonic gauge (SSPC-PA 9)	Each coat	DFT report
Consumption	Material used vs area	Each stage	Daily record
Continuity	Holiday test ASTM D4787 (concrete) / ASTM D5162 (steel)	100 % of lining	Holiday test report
Adhesion	Pull-off test on trial area / as specified (ASTM D7234 or D4541)	As specified	Test report
Final inspection	Appearance, continuity, finish, details	Each area	Handover checklist

15. Defects & Remedies

Defect	Likely cause	Remedy
Poor adhesion / delamination	Weak, dusty or damp substrate; recoat window exceeded	Remove, re-prepare and reapply; abrade and re-prime if window exceeded
Pinholes / bubbles	Outgassing; air trapped in mixing; thick application	Seal porous areas; mix slowly; spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash, abrade and recoat in suitable conditions
Soft spots	Incorrect ratio or incomplete mixing	Remove and reapply with complete kits
Holidays found	Insufficient thickness; pinholes	Abrade locally, recoat and re-test
Colour / shade variation	Different batches; uneven application	Use one batch per area; apply additional coat

16. Health, Safety & Environment

Hazard	Control measures
Skin / eye contact with resins and cement	Gloves, goggles, coveralls; eye-wash station
Inhalation of dust and vapours	Dust extraction; ventilation; masks / respirators
Fire (solvent vapours)	No smoking or hot work; fire extinguisher at the work area
Slips and trips	Housekeeping; barricades and warning signs

Hazard	Control measures
Manual handling	Correct lifting; mechanical aids for drums and pails
Electrical equipment	Inspected tools; protected cables with RCD
Confined spaces (tanks)	Permit to work, gas testing, forced ventilation, standby person

- Read the Safety Data Sheet of each product before use.
- Keep first-aid kit and eye-wash on site; follow the first-aid measures in each TDS.

16.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Allow small quantities of mixed resin to cure before disposal; dispose of according to local regulations.
- Do not discharge materials or washings into drains or soil.

17. Cleaning

- Clean tools of water-based and cementitious products with water immediately after use.
- Clean tools of epoxy and PU products with the recommended thinner before the material cures.
- Cured material can only be removed mechanically.

18. Limitations

- Do not apply outside the temperature and humidity limits stated in the TDS of each product.
- Do not apply on wet surfaces or when rain or dew is expected before the layer has cured.
- Do not mix or substitute products from other systems without written approval from O MIX.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Approved submittal and method statement on site	☐ Yes ☐ No	
Mock-up approved	☐ Yes ☐ No	
Substrate tests passed (strength, pull-off, moisture)	☐ Yes ☐ No	
Surface preparation approved	☐ Yes ☐ No	
Stage 1 — O-Prime ST applied and approved	☐ Yes ☐ No	
Stage 2 — O-Guard EP applied and approved	☐ Yes ☐ No	
Stage 3 — O-Guard EP applied and approved	☐ Yes ☐ No	
Recoat windows respected	☐ Yes ☐ No	
Final tests passed	☐ Yes ☐ No	
Area protected and handed over	☐ Yes ☐ No	

Inspected by	Signature	Date

Appendix B — Daily Application Record

Date	Area / zone	Stage	Batch No.	Qty used	Air / substrate °C	RH %	Sign

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O-Prime ST Data Sheet

O Poxyguard — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

O-Prime ST

Surface-Tolerant Epoxy Primer
for Steel, Light Metals & Concrete



Product Description

O-Prime ST is a two-component, solvent-based, surface-tolerant epoxy primer. It gives reliable adhesion on steel prepared by hand or power tools (St 2) as well as blast-cleaned steel (Sa 2½), and is suitable for concrete, light metals and their alloys. It is the base primer of the O MIX protective coating systems.

Uses

- Primer for structural steel, tanks, pipes and steel structures.
- Maintenance painting where only hand or power-tool cleaning is possible (St 2).
- Primer for light metals and their alloys.
- Primer on concrete before epoxy and polyurethane protective systems.
- Base coat in O MIX Acid Lock, Potable Water Tank and PU Protective systems.

Advantages

- Surface tolerant: St 2 or Sa 2½.
- High solids 85 %, high build.
- Excellent adhesion to steel and concrete.
- Wide DFT range 100 – 1000 µm.
- Compatible with epoxy and PU topcoats.

Application

1. **Surface Preparation**
   Remove rust, mill scale, oil and loose paint to St 2 minimum (Sa 2½ preferred).
2. **Mixing**
   Add Part B to Part A (4 : 1 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. **Application**
   Apply by brush, roller or airless spray at 0.18 – 0.20 kg/m² (100 µm DFT).

Consumption

0.18 – 0.20 kg/m²
at 100 µm DFT — theoretical 8.5 m²/L
Practical consumption depends on surface profile, method and losses.

Packaging

20 kg sets (Part A 16 kg + Part B 4 kg) and 5 kg sets (Part A 4 kg + Part B 1 kg) in metal pails.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight, heat and ignition sources.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	4 : 1 by weight	—
Finish	Semi-gloss	—
Colours	Grey, red, white	—
Solids content	85 %	—
Recommended DFT	100 – 1000 µm	—
Theoretical coverage	8.5 m²/L at 100 µm DFT	—
Pot life (25 °C)	2 hours	—
VOC	316 g/L	—
Flash point	27 °C	—
Min. substrate temperature	+10 °C	—
Shelf life	12 months	—

Consumption

DFT	Theoretical coverage	Consumption
100 µm	8.5 m²/L	0.18 – 0.20 kg/m²
150 µm	5.7 m²/L	0.27 – 0.30 kg/m²
200 µm	4.25 m²/L	0.36 – 0.40 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Substrate temperature: min. +10 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Steel: St 2 minimum; Sa 2½ for immersion or severe exposure.
- Concrete: at least 28 days old, sound, dry (< 4 % moisture) and free of laitance.

Application Notes

- Mix complete sets only; partial mixing must be weighed exactly at 4 : 1.
- Apply by brush, roller or airless spray; use brush for stripe coats on edges, welds and bolts.
- Overcoat with the specified O MIX epoxy or PU coat within the recoat window stated for the system.
- Thin only with the recommended epoxy thinner where needed for spray.
- Do not apply on wet surfaces or when condensation is expected.

Health & Safety

- Solvent-based: flammable. Keep away from heat, sparks and open flames; no smoking.
- Wear chemical-resistant gloves, safety goggles and overalls; use a respirator with organic vapour cartridges in enclosed spaces.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation during application and curing.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools and equipment immediately with epoxy thinner.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

Construction Chemicals

O-Guard EP Data Sheet

O Poxyguard — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

O-Guard EP

Solvent-Free Polyamine Epoxy Lining for Potable Water Tanks



Product Description

O-Guard EP is a solvent-free, polyamine-cured epoxy coating for lining potable water tanks and other concrete and steel structures. It is suitable for contact with drinking water and food, resists chemicals and corrosive vapours, and gives a glossy, easy-to-clean surface. An anti-slip finish is optional.

Uses

- Potable water tanks and reservoirs.
- Chemical and wastewater tanks.
- Industrial concrete and steel tanks.
- Car parks and bridge decks.
- Pharmaceutical and textile plants, basements and power stations.

Advantages

- Suitable for potable water contact.
- Solvent-free, 100 % solids.
- Resists chemicals and corrosive vapours.
- Glossy, easy to clean.
- Can be used in lamination systems.

Application

1. **Surface Preparation**
   Concrete ≥ 25 N/mm², pull-off ≥ 1.5 N/mm², ≥ 28 days, moisture < 4 %; prime with O-Prime ST.
2. **Mixing**
   Add Part B to Part A (200 : 30 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. **Application**
   Apply by roller or airless spray at 0.25 – 0.30 kg/m² per coat; second coat within 48 hours.

Consumption

0.25 – 0.30 kg/m²
per coat (approx. 200 µm)
Two coats for tank linings.

Packaging

5 kg sets: Part A 4.35 kg in pail + Part B 0.65 kg in jerrycan.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	200 : 30 by weight — pre-weighed sets, mix as recommended	—
Specific gravity	1.57	—
Solids content	100 %	—
Flash point	145 °C	—
Pot life	40 min	—
Recoat	24 hours (max. 48 hours)	—
Full cure	7 days	—
Heat resistance	up to 80 °C	—
Colours	On request	—

Consumption

Layer	Product	Consumption
Primer	O-Prime ST	0.18 – 0.25 kg/m²
Body coat	O-Guard EP	0.25 – 0.30 kg/m²
Top coat	O-Guard EP	0.25 – 0.30 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Concrete ≥ 25 N/mm², pull-off ≥ 1.5 N/mm², ≥ 28 days, moisture < 4 %.

Application Notes

- Mix complete sets.
- Apply the second coat within 48 hours.
- Allow 7 days full cure and wash the tank with clean water before filling with potable water.
- Broadcast fine quartz sand on floors where anti-slip is required.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with the recommended thinner before the material cures.

Construction Chemicals

Test Reports

O Poxyguard — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheets.

Construction Chemicals

Approvals & Certificates

O Poxyguard — Submittal Document

To be attached: approvals and certificates, where available.